Chapter 14 – Database and Database Management Systems

GE-161 Introduction to Information and Communication Technologies

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Learning Objectives

1. Explain what a database is, including common database terminology, and list some of the advantages and disadvantages of using databases.
2. Discuss some basic concepts and characteristics of data, such as data hierarchy, entity relationships, and data definition.
3. Describe the importance of data integrity, security, and privacy, and how they affect database design.

Learning Objectives

4. Identify some basic database classifications and discuss their differences.
5. List the most common database models and discuss how they are used today.
6. Understand how a relational database is designed, created, used, and maintained.
7. Describe some ways databases are used on the Web.

Overview

- This chapter covers:
  - What a database is, the individuals who use them, and how databases evolved
  - Important database concepts and characteristics
  - Database classifications and models
  - How to create and use a relational database
  - How databases are used on the Web

What Is a Database?

- Database: A collection of related data stored in a manner so it can be retrieved as needed
- Database management system (DBMS): Used to create, maintain, and access databases
- A database typically consists of:
  - Tables: Collection of related records
  - Fields (columns): Single category of data to be stored in a database (name, telephone number, etc.)
  - Records (rows): Collection of related fields in a database (all the fields for one customer, for example)
- Relational database: Data from several tables is tied together (related) using a field that the tables have in common

A Simple Relational Database Example

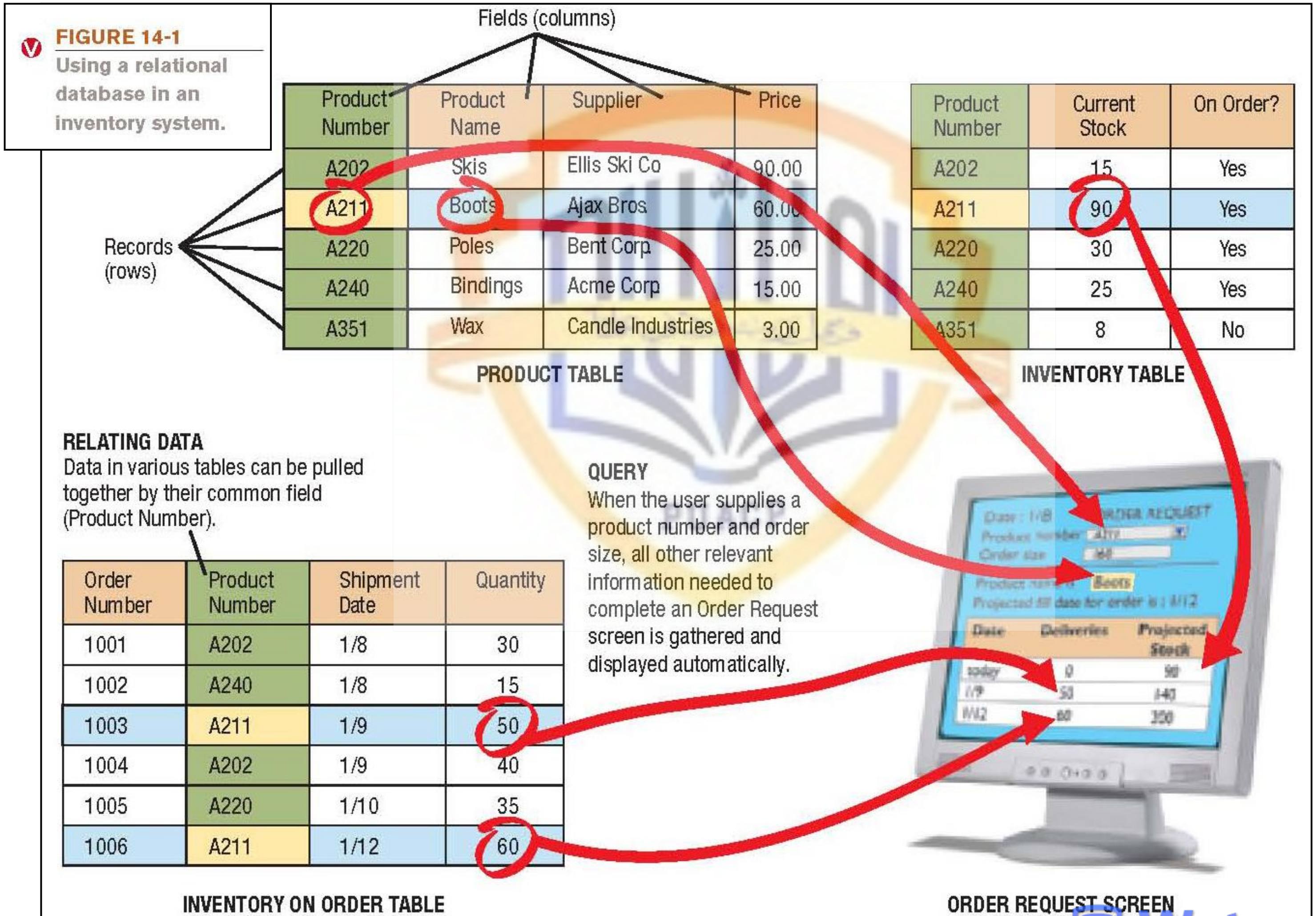


FIGURE 14-1
Using a relational database in an inventory system.

PRODUCT TABLE

Product Number	Product Name	Supplier	Price
A202	Skis	Ellis Ski Co	90.00
A211	Boots	Ajax Bros	60.00
A220	Poles	Bent Corp	25.00
A240	Bindings	Acme Corp	15.00
A351	Wax	Candle Industries	3.00

INVENTORY TABLE

Product Number	Current Stock	On Order?
A202	15	Yes
A211	90	Yes
A220	30	Yes
A240	25	Yes
A351	8	No

INVENTORY ON ORDER TABLE

Order Number	Product Number	Shipment Date	Quantity
1001	A202	1/8	30
1002	A240	1/8	15
1003	A211	1/9	50
1004	A202	1/9	40
1005	A220	1/10	35
1006	A211	1/12	60

What Is a Database?

- Primary key: Specific field that uniquely identifies the records in that table
  - Used in a relational database to relate tables together
  - Must be unique and a field that doesn't change
- PC DBMSs include:
  - Microsoft Access, Corel Paradox, Lotus Approach
- For more comprehensive enterprise databases
  - Oracle Database, IBM DB2, Microsoft SQL Server

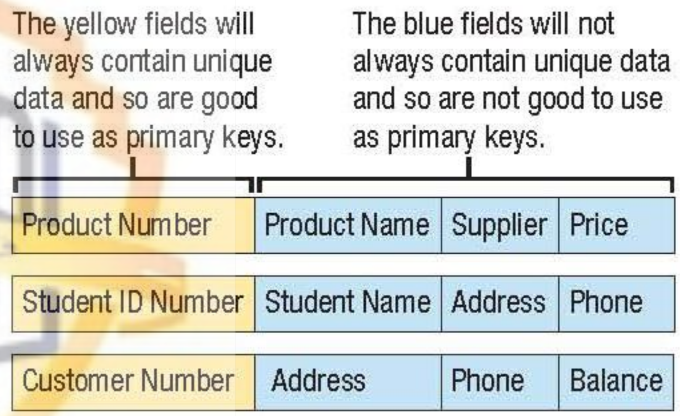


FIGURE 14-2
Key fields. A key field must contain unique data so it can be used to identify each record in the table.

What Is a Database?

- Individuals involved with a DBMS:
  - Database designers: Design the database
  - Database developers: Create the database
  - Database programmers: Write the programs needed to access the database or tie the database to other programs
  - Database administrators: Responsible for managing the databases within an organization
  - Users: Individuals who enter data, update data, and retrieve information out of the database

The Evolution of Databases

MODEL	FLAT FILES	HIERARCHICAL	NETWORK	RELATIONAL	OBJECT-ORIENTED	MULTI-DIMENSIONAL
YEAR BEGAN	1940s	1960s	1960s	1970s	1980s	1990s
DATA ORGANIZATION	Flat files	Trees	Trees	Tables and relations	Objects	Data cubes, tables and relations, or a combination
DATA ACCESS	Low-level access	Low-level access with a standard navigational language	Low-level access with a standard navigational language	High-level, nonprocedural languages	High-level, nonprocedural, object-oriented languages	OLAP tools or programming languages
SKILL LEVEL REQUIRED TO ACCESS DATA	Programmer	Programmer	Programmer	User	User	User
ENTITY RELATIONSHIPS SUPPORTED	One-to-one	One-to-one, one-to-many	One-to-one, one-to-many, many-to-many	One-to-one, one-to-many, many-to-many	One-to-one, one-to-many, many-to-many	One-to-one, one-to-many, many-to-many
DATA AND PROGRAM INDEPENDENCE	No	No	No	Yes	Yes	Yes

FIGURE 14-3

The evolution of databases. Databases have evolved over the years, becoming more flexible, more capable, and easier to use.

Advantages and Disadvantages of the DBMS Approach

- Advantages
  - Faster response time
  - Lower storage requirements
  - Easier to secure
  - Increased data accuracy
- Disadvantages
  - Increased vulnerability (backup is essential)

Data Concepts and Characteristics

- Data hierarchy
  - Characters
  - Fields/columns: Hold single pieces of data
  - Records/rows: Groups of related fields
  - Tables: Collection of related records
  - Database: Contains a group of related tables
- Entity: Something of importance to the organization
  - Entities that the organization wants to store data about typically becomes a database table
  - Attributes: Characteristics of an entity
    - Typically become fields in the entity's database table

Data Concepts and Characteristics

- Entity relationships: Describe an association between two or more entities
  - One-to-one (1:1) entity relationships (not common)
    - e.g. each store has a single manager
  - One-to-many (O:M) entity relationships (most common)
    - e.g. a supplier supplies more than one product to a company
  - Many-to-many (M:M) entity relationships (requires a third table to tie the tables together)
    - e.g. an order can contain multiple products and a product can appear on multiple orders

Data Concepts and Characteristics

- Data definition: The process of describing the properties of data to be included in a database table
  - During data definition, each field is assigned:
    - Name (must be unique within the table)
    - Data type (such as Text, Number, Currency, Date/Time)
    - Description (optional description of the field)
    - Properties (field size, format of the field, allowable range, if field is required, etc.)
- Finished specifications for a table become the table structure

Data Definition

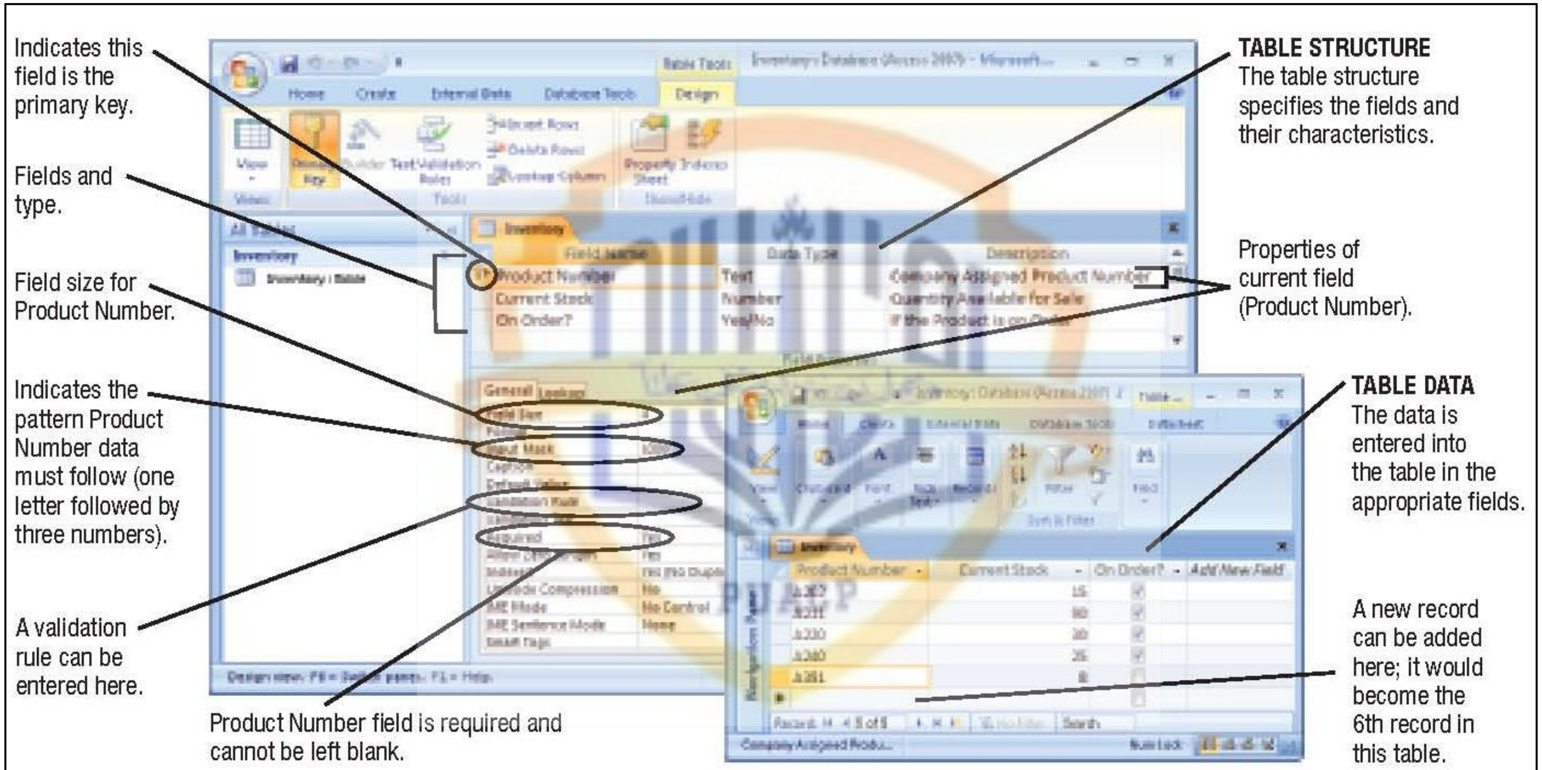


FIGURE 14-4

Data definition. Each field in a database has a defined data type and properties that can be assigned to that field.

Data Concepts and Characteristics

- Data dictionary: Contains all data definitions in a database
  - Table structures
    - Names, types and properties of each field
  - Security information (passwords, etc.)
  - Relationships between the tables in the database
  - Current information about each table, such as the current number of records
  - Does not contain any of the data in the tables
    - Metadata: Data about the database tables
  - Ensures that data being entered into the database does not violate any specified criteria

Data Concepts and Characteristics

- Data integrity: The accuracy of data
  - Quality of data input determines the quality of retrieved information
  - Data validation: Ensuring that data entered into the database is valid
    - Record validation rules: Checks all fields before changes to a record are saved
    - Can be enforced on a per transaction basis so the entire transaction will fail if one part is invalid
  - Database locking
    - Prevents two individuals from changing the same data at the same time

Data Validation

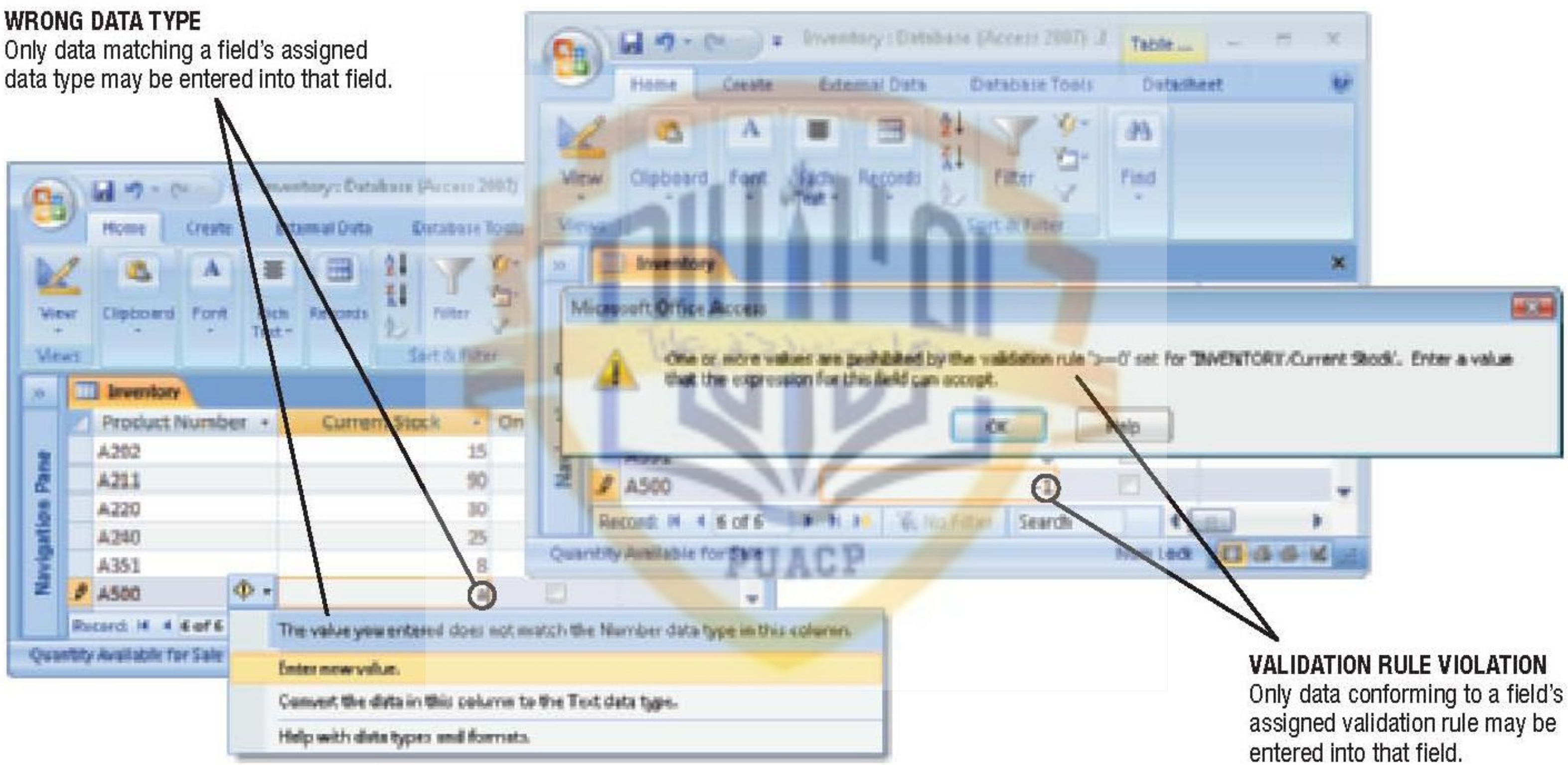


FIGURE 14-5

Data validation.

Good data validation rules can prevent invalid data from being entered into a database table.

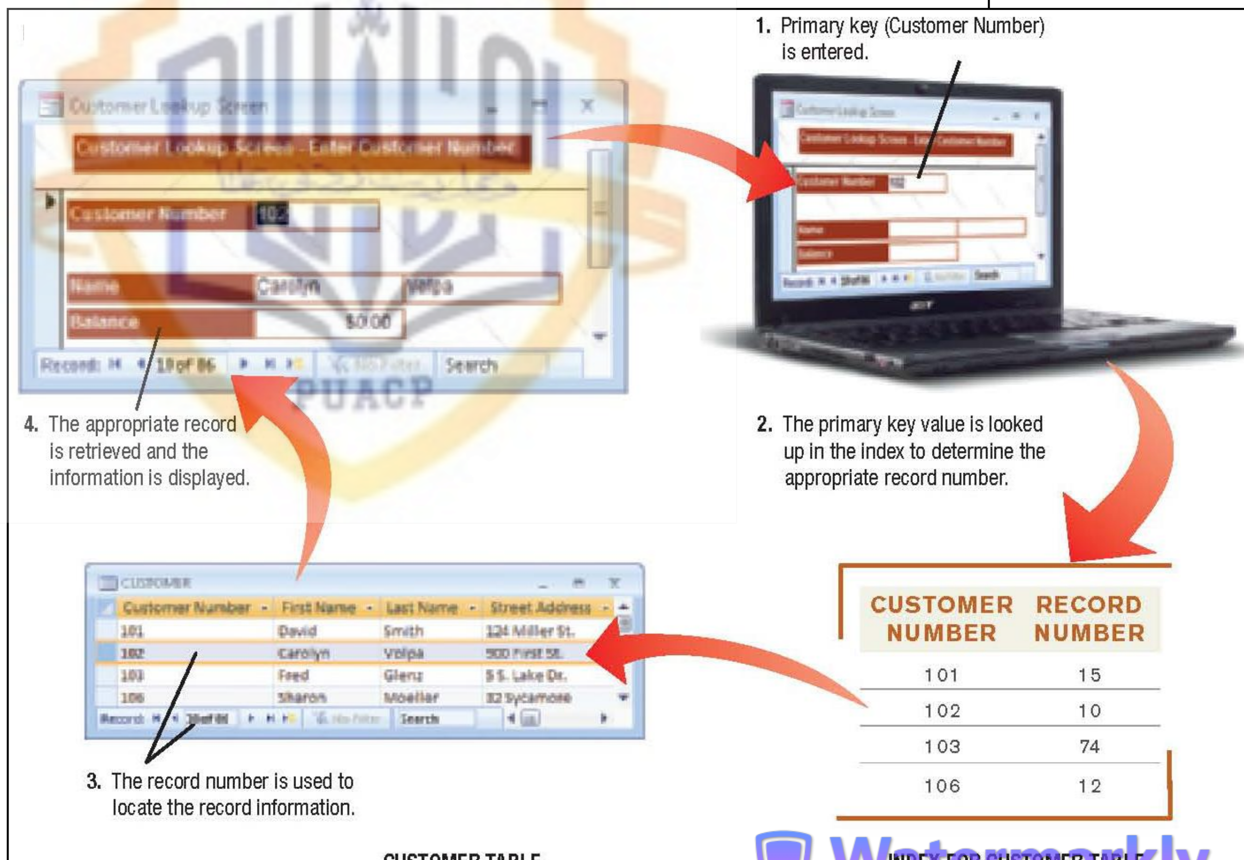
Data Concepts and Characteristics

- Data security: Protecting data against destruction and misuse
  - Protects against unauthorized access
  - Database activity monitoring programs can be used to detect possible intrusions
  - Should include strict backup and disaster-recovery procedures (disaster-recovery plan)
    - Protects against data loss
- Data privacy: Growing concern because of the vast amounts of personal data stored in databases today

Data Concepts and Characteristics

- Data organization: Arranging data for efficient retrieval
  - Indexed organization uses an index to keep track of where data is stored

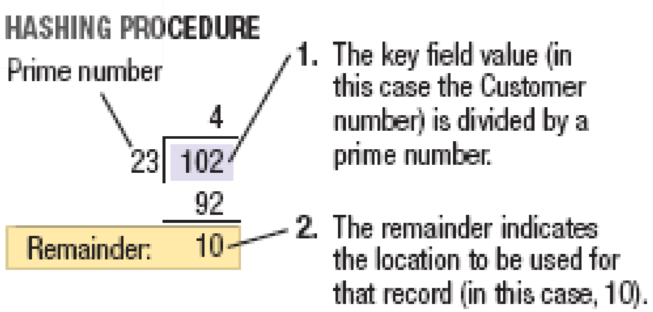
FIGURE 14-7
Indexed organization is often used for real-time transaction processing.



Data Concepts and Characteristics

– Direct organization
  - Uses hashing algorithms to specify the exact storage location
  - Location is based on primary key
  - Algorithms should be designed to limit collisions
– Sometimes a combination of indexing and direct organization is used within a database system

FIGURE 14-8 Direct organization is frequently used for faster real-time processing.



Quick Quiz

1. A column in a database in which customer names are stored would be referred to as a ________________________.
   a. field
   b. record
   c. table
2. True or False: Data validation procedures are used to ensure that data entered into a database matches the specified type, format, and allowable value.
3. The ________________________ contains metadata about the database tables in a database.

Answers:

1) a; 2) True; 3) data dictionary

Database Classifications

- Single-user database system: Designed to be accessed by one user
- Multiuser database system: Designed to be accessed by multiple users (most business databases today)
- Client-server database systems: Has both clients (front end) and at least one database server (back end)

FIGURE 14-9
Client-server database systems.

FRONT END (CLIENT COMPUTERS)
The client computers typically utilize a graphical user interface to access the database located on the back-end server.

BACK END (SERVER)
The server contains the database used to fulfill the requests of the client computers.

Database Classifications

- N-tier database system: Has more than two tiers
  - Middle tiers contain one or more programs stored on one or more computers
    - Program code is separate from the database
    - Provides flexibility and scalability

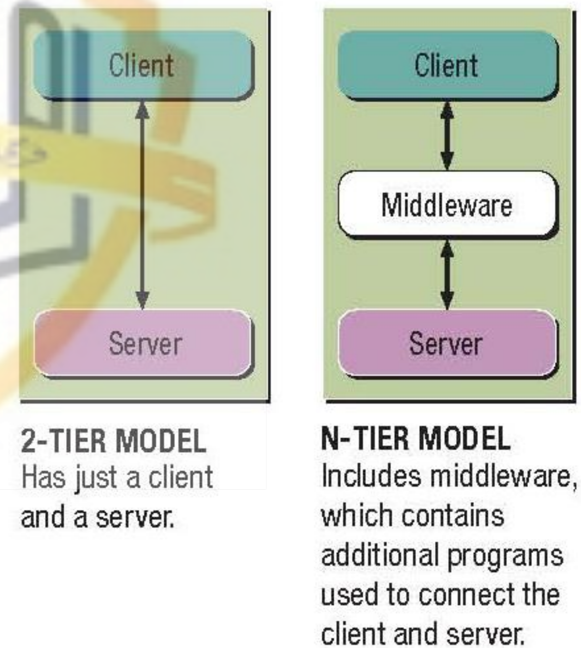
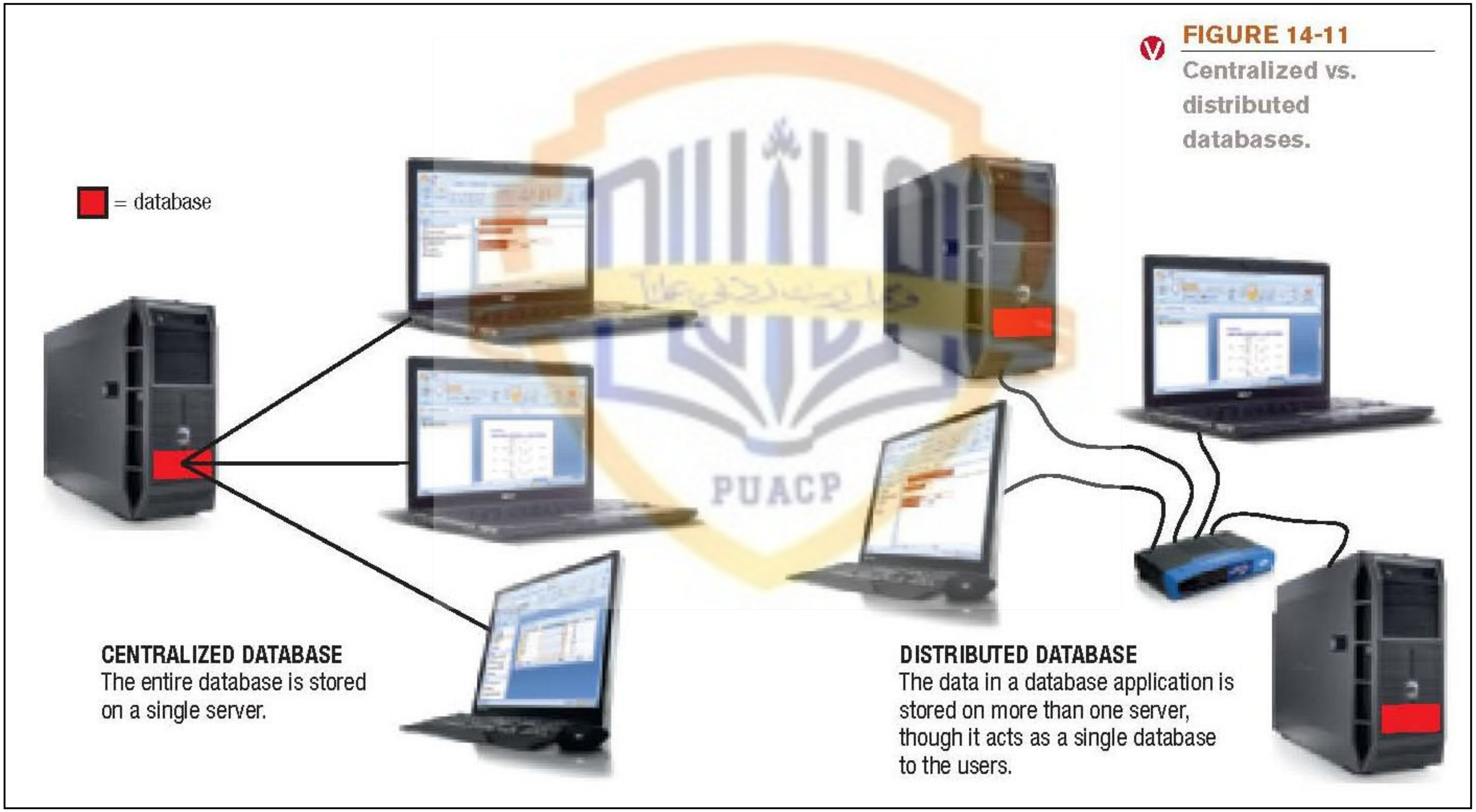


FIGURE 14-10
A 2-tier vs. an n-tier database model.

Database Classifications

- Centralized database system: Database is located on a single computer, such as a server or mainframe
- Distributed database system: Data is physically divided among several computers connected by a network, but the database logically looks like it is a single database
- Disk-based databases: Data is stored on hard drives
- In-memory databases (IMDBs): Data is stored in main memory
  - Faster, used when performance is critical
  - Good backup procedures are essential

Database Classifications



Quick Quiz

1. Which type of database system is beginning to be used in high-end systems where performance is crucial?
   a. In-memory databases
   b. Disk-based databases
   c. Single-user databases
2. True or False: With the n-tier database model, there is at least one middle piece of software between the client and the server.
3. With a(n) ________________________ database system, the databases used by the system are all located on a single computer.

Answers:

1) a; 2) True; 3) centralized

Database Models

- Two older models:
  - Hierarchical databases: Store data in the form of a tree, with typically a one-to-many relationship between data entities
  - Network databases: Show the relationship between data elements usually as either one-to-many or many-to-many
- Relational database management system (RDBMS)
  - Data is organized in tables related by common fields
  - Most widely used database model today

The Relational Database Model

- Database design steps
  - Identify the purpose of the database
  - Determine the tables and fields
  - Assign the fields to a table and reorganize as needed to minimize redundancy (normalization)
  - Finalize the structure (primary keys, field properties, etc.)

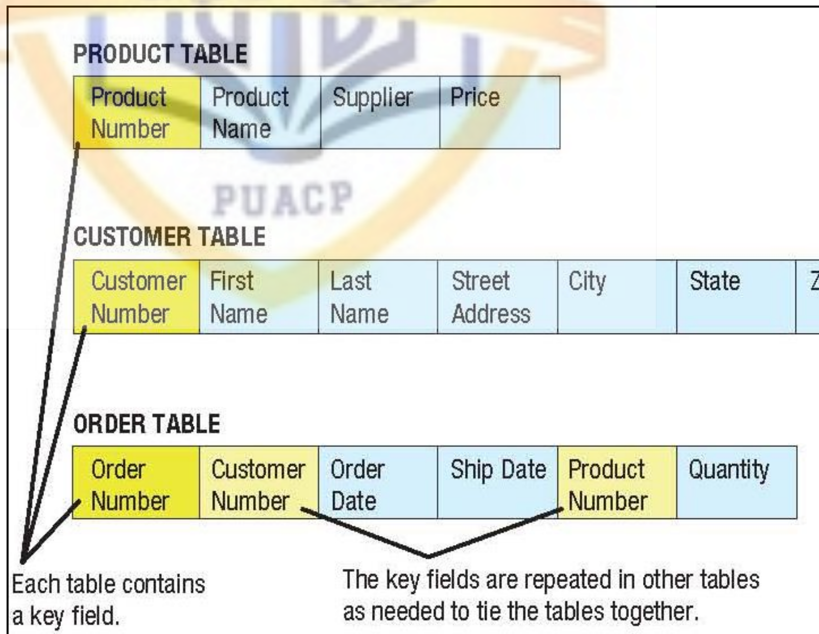


FIGURE 14-13 A preliminary design for three tables in the Inventory database.

The Relational Database Model

- Creating a relational database:
  - Create the database file
  - Create the structure of each individual table (in Access, can be performed in either Design or Datasheet view)
  - Enter data
    - Existing data can be migrated to the new database
    - New data can be added via form or the Design view
  - Relate tables as needed

The Relational Database Model

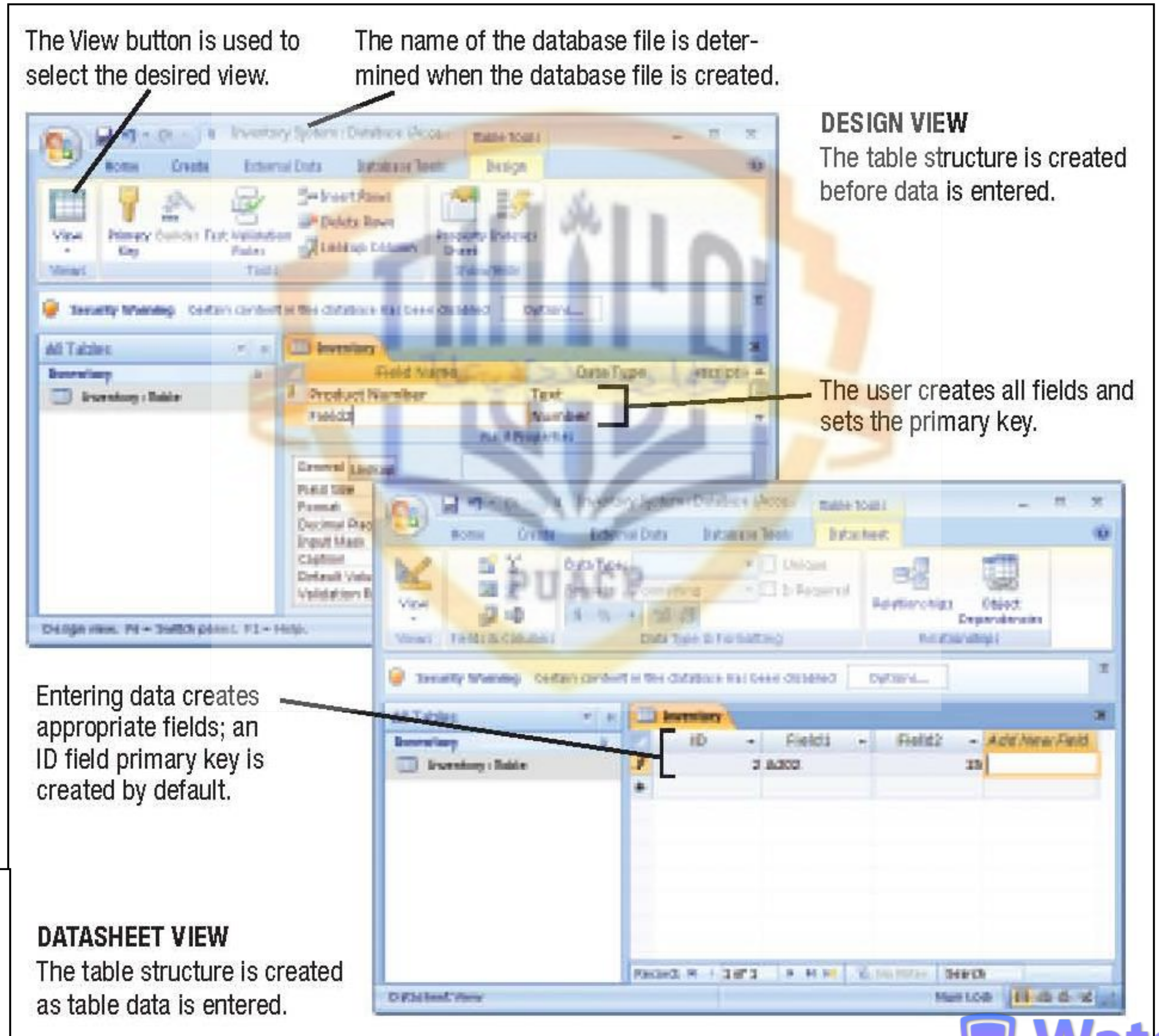


FIGURE 14-14 Tables can be created using Design or Datasheet view.

The Relational Database Model

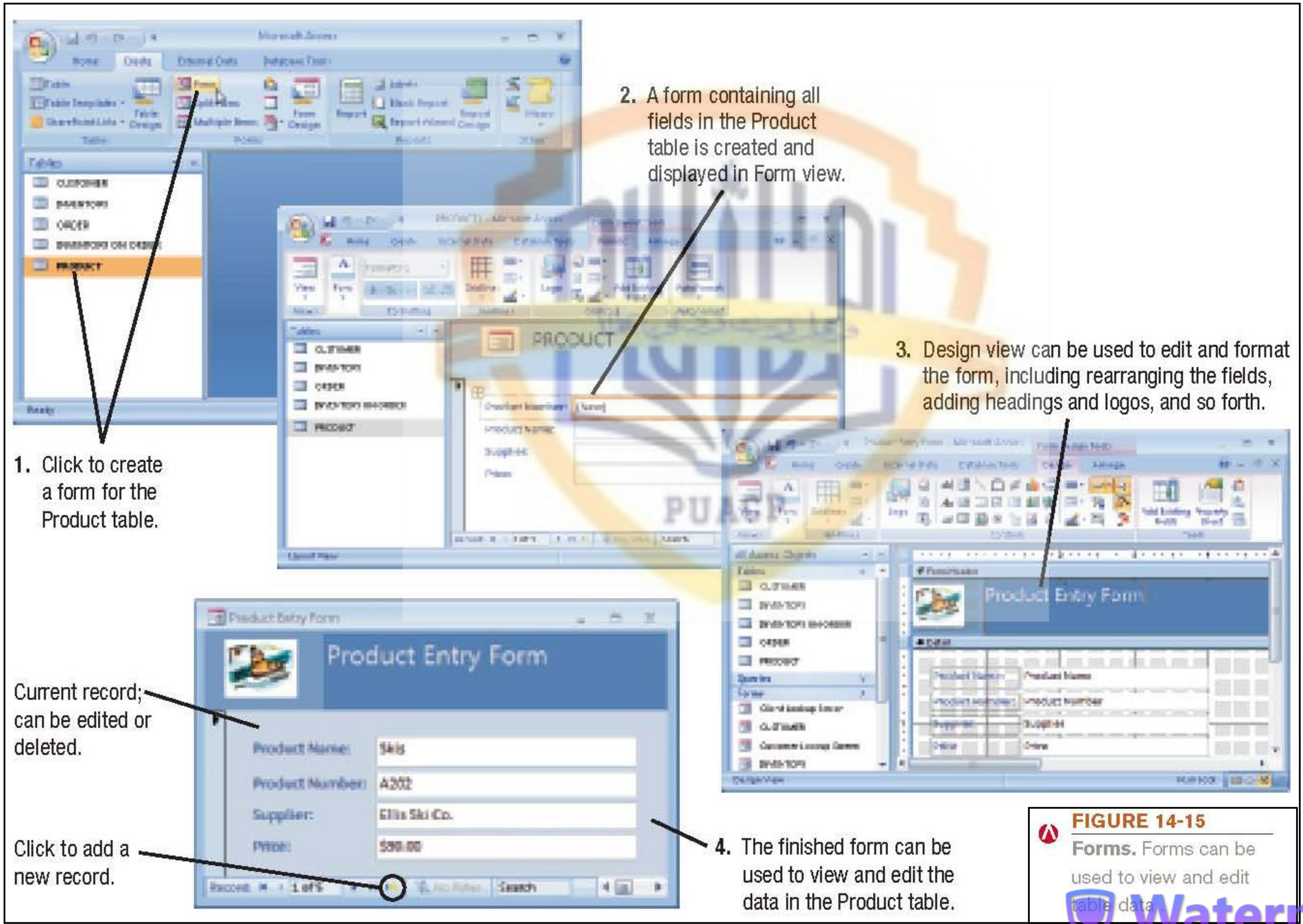


FIGURE 14-15
Forms. Forms can be used to view and edit table data.

The Relational Database Model

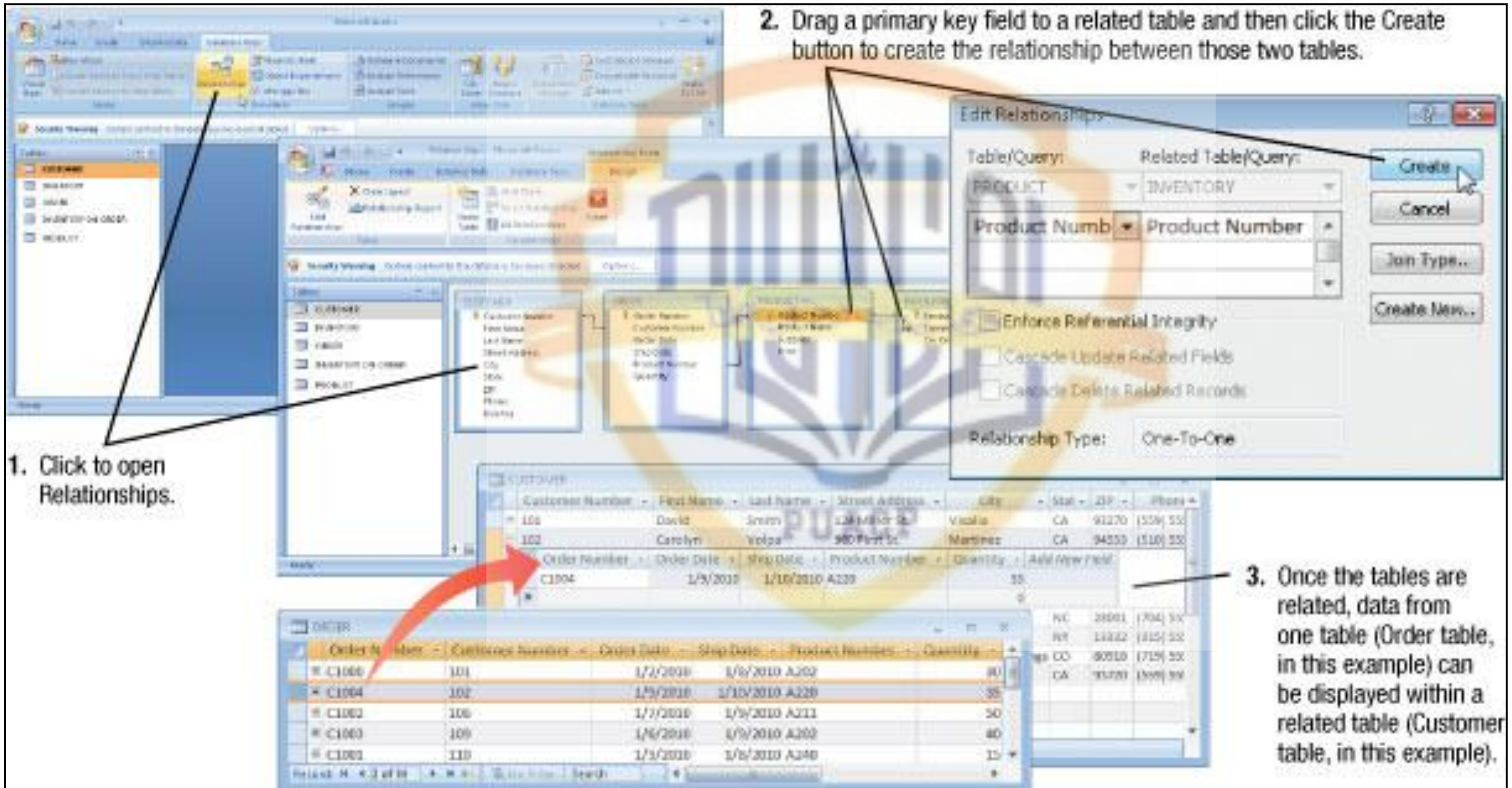


FIGURE 14-16
Relating tables.

The Relational Database Model

- Retrieving information from database
  - Query: A request to see information from a database that matches specific criteria
    - Specifies which records should be retrieved by specifying criteria
    - Can specify the fields to be displayed
    - Many programs have wizards or other tools to make it easy to create a query
    - Must be designed to extract information as efficiently as possible
    - Queries are saved so they can be retrieved again when needed; proper results are displayed each time the query is run

The Relational Database Model

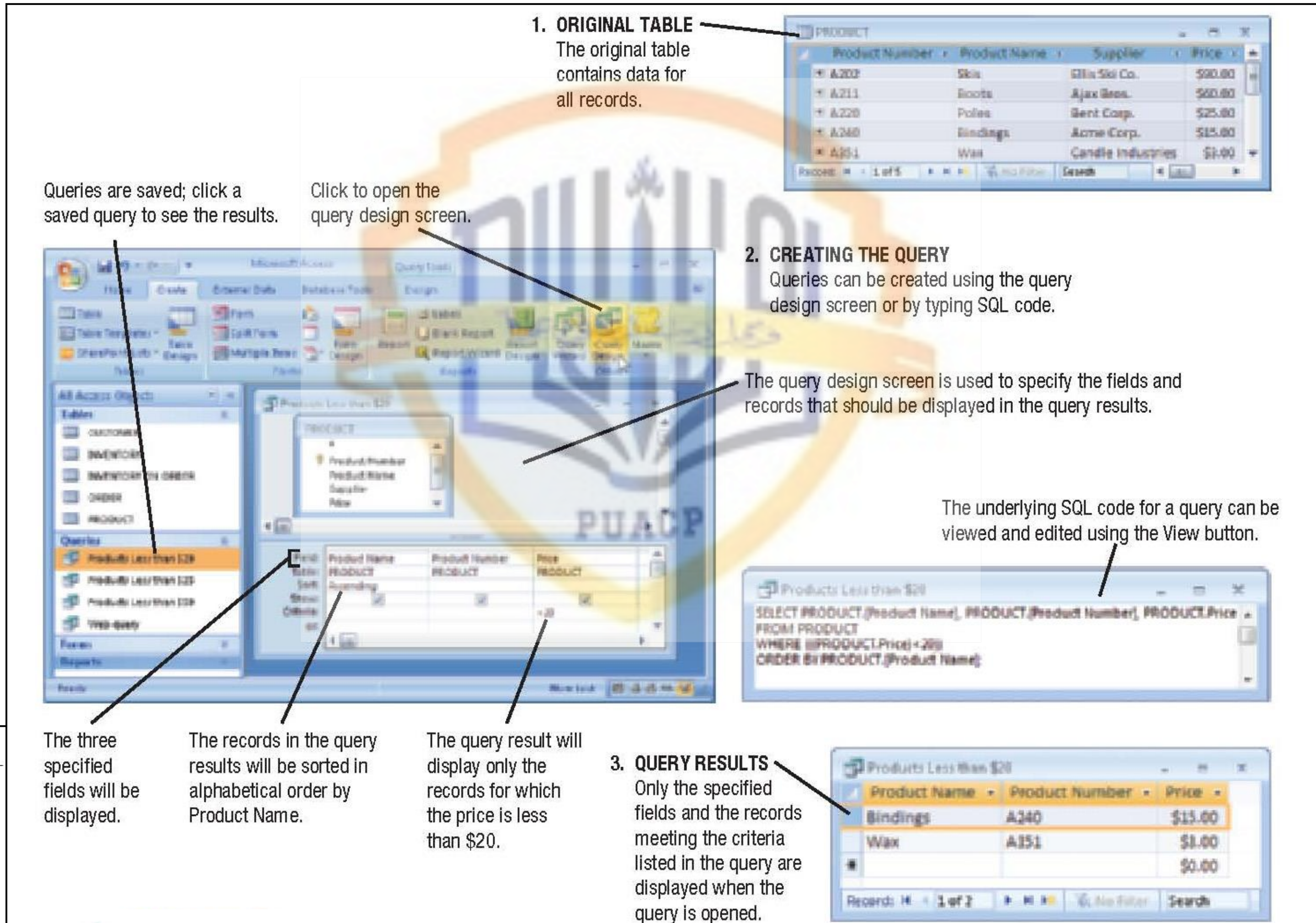


FIGURE 14-17
Querying a database. This example pulls information from the Product table in the Inventory database.

The Relational Database Model

– Report: Formatted means of looking at a database table or the results of a query
  - Reports can pull data from more than one table
  - Many programs have wizards or other tools to make it easy to create a report
  - Can be modified and customized using the Design view
  - Reports are saved so they can be retrieved again when needed; proper results are displayed each time the query is run

The Relational Database Model

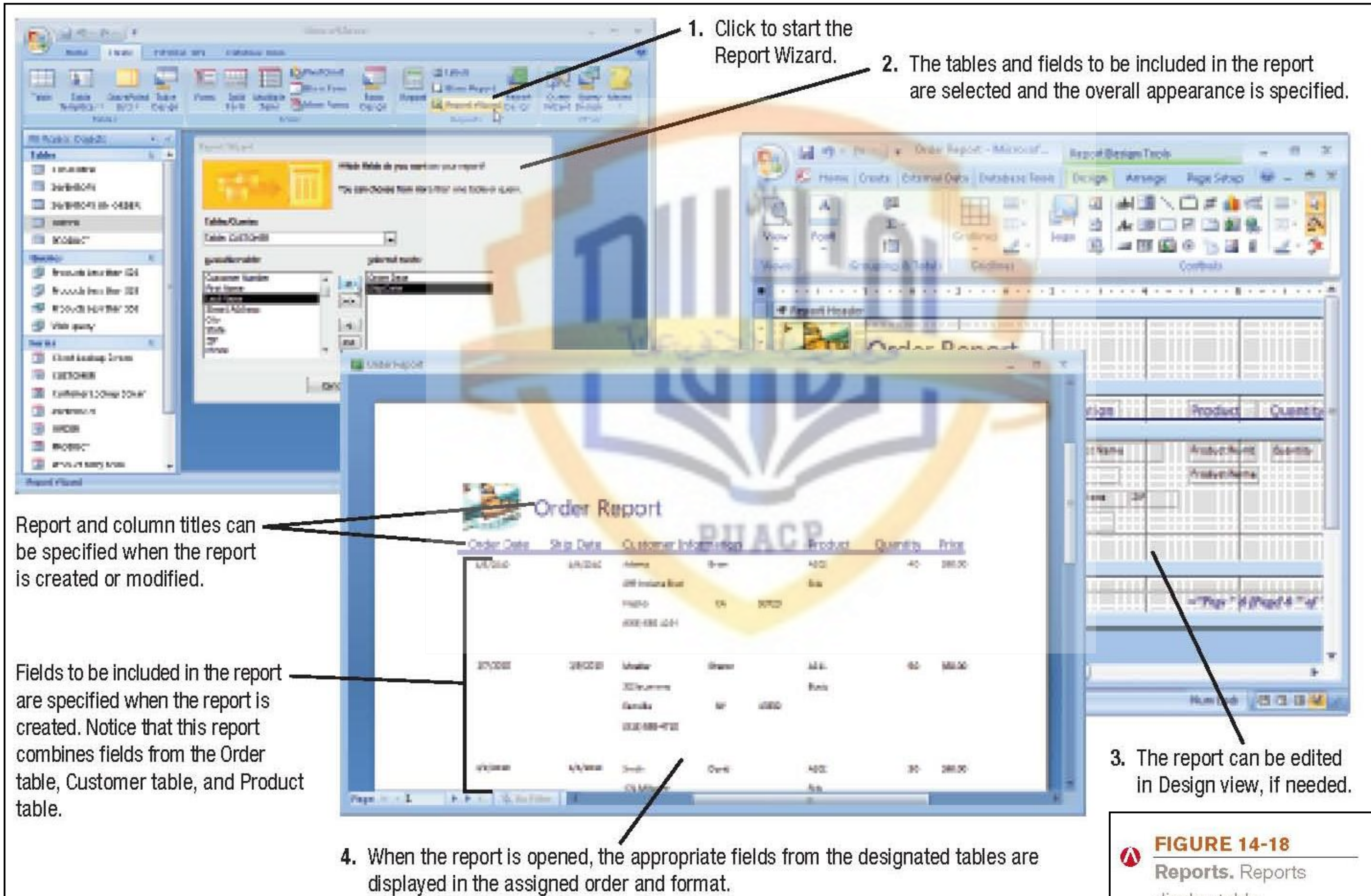


FIGURE 14-18
Reports. Reports display table information with a more formal, businesslike appearance.

The Relational Database Model

- Maintaining a database
  - Data in tables can be edited as needed
  - Table structures can be modified when needed
  - Other possible modifications:
    - Adding new indexes to speed up queries
    - Deleting obsolete data
    - Upgrading database software, installing patches
    - Repairing/restoring data that has become corrupt
    - Continuing to evaluate and improve security

The Object-Oriented Database Model

- Object-oriented database management system (OODBMS): Database system in which multiple types of data are stored as objects along with their related code
  - Objects contain data along with the methods that can be taken with that data
  - Objects in an OODBMS can contain virtually any type of data—video clip, photograph with a narrative, text with music, and so on—along with the methods to be used with that data
  - Objects can be retrieved using queries
    - OQL – Object Query Language
      - OO version of SQL

The Object-Oriented Database Model

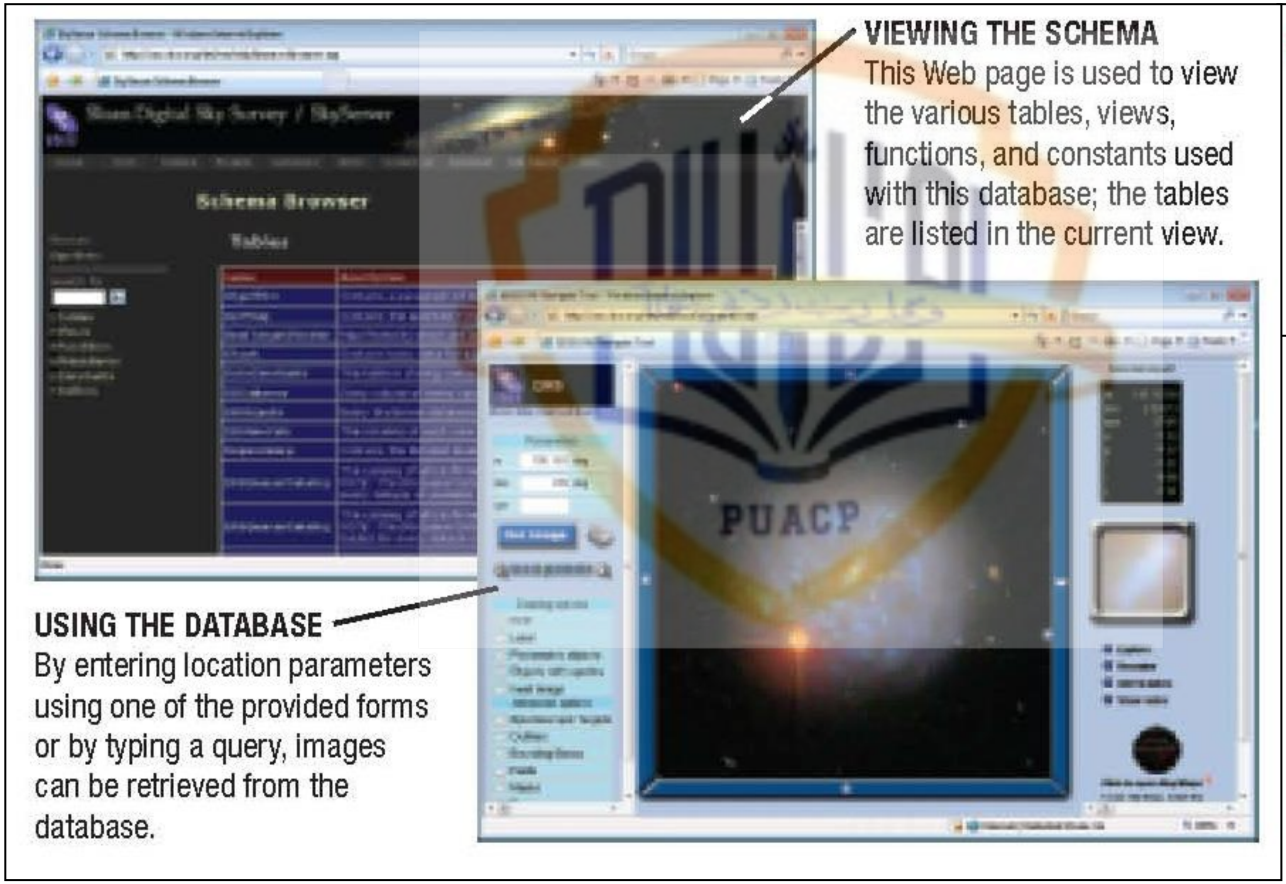


FIGURE 14-19

Object-oriented databases. The database shown here contains images and information from the Sloan Digital Sky Survey.

Hybrid Database Models

- Hybrid database: A combination of database types or models
  - Hybrid XML/relational database: Can store and retrieve both XML data and relational data

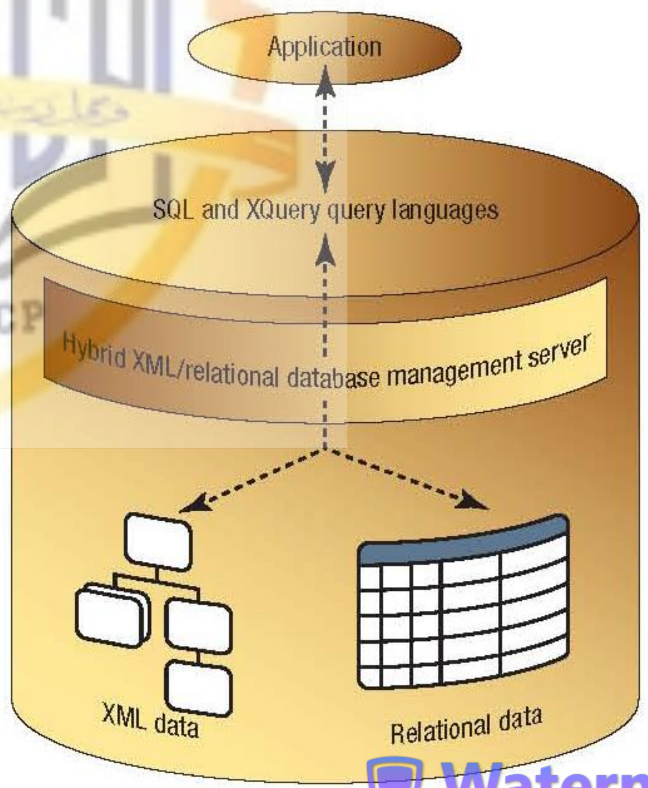


FIGURE 14-20
Hybrid XML/relational databases.

Multidimensional Databases

- Multidimensional database (MDDB): Type of database designed to be used with data warehousing
  - Often used in conjunction with Online Analytical Processing (OLAP)
    - MOLAP (Multidimensional OLAP): Data is stored in single structures called data cubes
    - ROLAP (Relational OLAP): Data is stored in an existing relational database using tables to store the summary information
    - HOLAP (Hybrid OLAP): Combination of MOLAP and ROLAP technologies

Databases and the Web

- Databases are commonly used on the Web
  - Information retrieval, e-commerce, dynamic Web pages (change based on user input), etc.

REFERENCE SITE
This site stores address information for individuals and businesses in the United States. After the user enters a name or category and a location (in this case, pizza parlors in Pismo Beach, California), the matching information is retrieved from the database.

PERSONALIZED SITE
This site retrieves information from its database to create a personalized page for signed-in viewers, such as recently viewed products and recommendations based on viewed and purchased products.

FIGURE 14-21
Database use on the Web.

Databases and the Web

- How Web databases work
  - Visitor makes request via a Web site
    - Search form
    - Logging on to personalize site
    - Uploading user content
  - Web server converts the request into a database query and passes it onto the database server, and then sends the results back to the visitor

Databases and the Web

- Middleware is used to connect two otherwise separate applications, such as a Web server and a database management system
  - Commonly written as scripts
    - JavaScript
    - VB Script
    - CGI scripts
    - Active Server Pages (ASPs)
    - PHP scripts

Databases and the Web

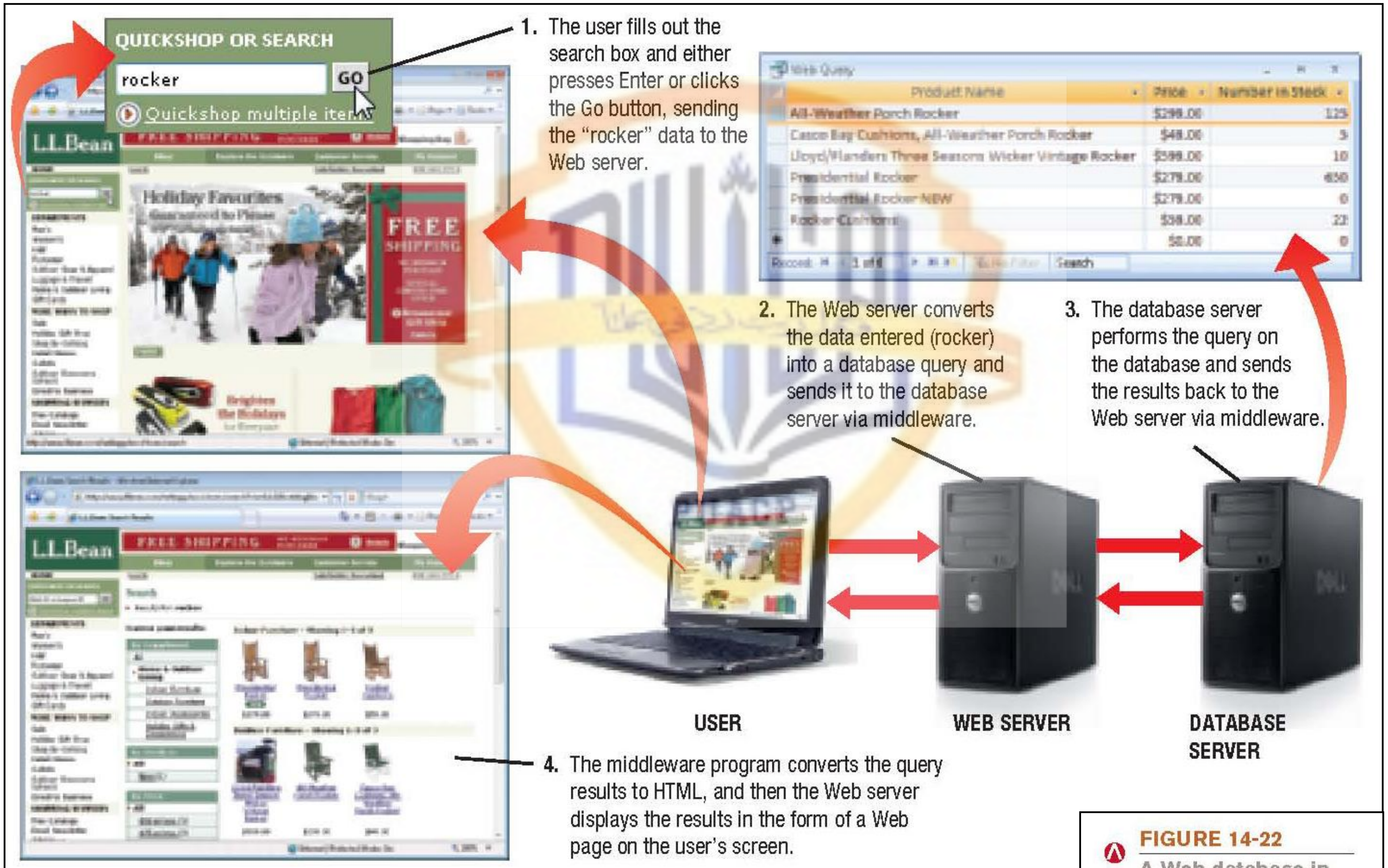


FIGURE 14-22
A Web database in action.

Quick Quiz

1. Which of the following is the most widely used type of database today?
   a. Network
   b. Relational
   c. Object-oriented
2. True or False: Databases are often used in conjunction with dynamic Web pages.
3. A(n) ________________________ is used to extract specific information from a database by specifying particular conditions about the data to be retrieved.

Answers:

1) b; 2) True; 3) query

Summary

- What Is a Database?
- Data Concepts and Characteristics
- Database Classifications
- Database Models
- Databases and the Web